

ALVIN RACHMAT

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Work Experience

ARTEFACT CONSULTING SINGAPORE PTE LTD

*Artefact is a global data science & AI consulting firm headquartered in France, with an international presence including Southeast Asia. Alvin specializes in **banking-domain AI & RAG-chatbot systems**.*

Data Scientist

11/25 – Now

Abu Dhabi, UAE (On-Site)

- Maintaining data governance workflows for the Abu Dhabi Department of Government Enablement through **LLM-assisted** rule generation & quality checks across **~800 datasets**
- Built an **entity resolution & deduplication** pipeline using Splink's unsupervised probabilistic record linkage to match and merge citizen records across government tables, enriching incomplete records (e.g. backfilling missing emails via shared national ID)
- Built an **LLM-powered** remediation agent on **Databricks using LangChain** that scans tables against governance rules and generates proposed corrections as output tables for steward review

Philippines (On-Site)

- **Sales leads (PoS) optimization** project with **RCBC Bank** (5th largest private bank in the Philippines), using Snowflake, Google Maps API, and geospatial visualization (Folium) to identify and prioritize high-potential business locations
- Built **data-driven lead scoring and mapping solutions, including Shortest Route Problem & Clustering**, enabling commercial teams to improve targeting efficiency and on-ground execution to **70% lead acceptance rate**
- Leveraged **AI heavily (Claude Code, Cursor, Databricks Genie, Snowflake Cortex)** to improve project delivery speed without sacrificing quality, validating all AI output through hands-on review

Indonesia (On-Site)

- Delivered **end-to-end AI solutions (Speech-to-Text, RAG, LLM evaluation)** for **CIMB Bank** (2nd largest private bank in Indonesia by assets), to support customer service transformation
- Architected a RAG pipeline on Speech-to-Text outputs and owned the evaluation strategy, improving **precision from 20% to 70% and recall from 20% to 85%**, achieved **NPS of 50%** among CS agents
- **Stack:** Whisper, LangChain, Elasticsearch (hybrid retrieval), Redis, FastAPI

UNTUKMU.AI & FORU AI

Joined Untukmu as ML Engineer building AI gifting chatbot (rebuilt, self-hosted 2026 at dewi.alvinrach.my.id), recommendation engine, & B2B integrations. Transitioned to ForuAI in mid-2024, developing its decentralized AI platform: AI-identity agents, data-monetization pipelines, & the supporting ML infrastructure.

Machine Learning Engineer

09/23 – 11/25

1. **LLM Development:** Utilized **LLM Models** to empower AI Chatbot functionality, enhance **process efficiency** through concurrency, parallelism & prompt engineering (Bahasa & English), resulting in **faster response time** — **Retraining** data from OpenAI to **reduce prompt cost** — Developing **metrics for AI chatbot** quality & the testing framework — **Research** Edge AI & Inverse Embedding **Paper** — Research & deploy **local/on-premise LLM Model** — **Training/Fine-tuning LLM** to a dataset to match the question we want (BERT, LLaMa).
 - ↳ **Stack:** FastAPI & Flask, OpenAI (Text Gen, Image Gen, Embeddings), Gemini, LM Studio, Ollama, Autogen, HuggingFace, Langchain, PyTorch
- **MLOps & Cloud Infra:** Managed **daily Google Cloud** operations: VM, serverless, Kubernetes, storage, CI/CD & alert systems.

- ↳ **Stack:** Compute Engine, Cloud Run, Kubernetes Engine, Bucket, Trigger, Cloud Build & Artifact, Vertex, Cloud Function, Logging Sink, Pub/Sub, Microsoft Azure (GenAI only)
- **Search & Retrieval Engine:** Developed **hybrid search** pipelines between **BM25 & semantic search** (vector database).
 - ↳ **Stack:** Elasticsearch, Milvus, ChromaDB
- **AI Apps Development & Interfaces:** Develop **Telegram Bot** for auto-translation of posts to each community — Develop **AI Testing Framework** with user-friendly UI for efficient chatbot testing.
 - ↳ **Stack:** Python's Telebot, Streamlit, HTML/CSS
- **External Integrations & Specialized AI Projects:** **Agentic LLM** using a new specialized agentic blockchain-socmed framework (Eliza - **Node JS**) — Working with **Somnia based API (blockchain)** to review referral NFT count of the address — Helping BE to integrate **Social Media API** (Instagram API) — Deploy faceswap technology using open source packages (**Computer Vision**).
 - ↳ **Stack:** Eliza (NodeJS), Faceswap, Instagram API, Somnia API
- **Leadership, Prototyping & Presentations:** Become **AI team PIC**: Manage AI team, handle interview, & mentored two junior team members — **Rapid development** in prototyping to investors (web dashboard and usability of the LLM apps)

SINAR MITRA SEPADAN FINANCE (Part of Orix Japan and Sinarmas)

A national Indonesian finance company, subsidiary of ORIX Corporation and Sinarmas Multiartha, specializing in used car and multipurpose consumer financing services.

Data Scientist

05/22 – 09/23

- Developed **credit scoring models** in collaboration with cross-functional teams, including BOD members, credit analysts, product manager, data engineer, data scientist, and backend engineer
- **Presented** credit scoring model results to BODs from Japan & Indonesia on a monthly basis
- Collaborated with collection team to develop a predictive model to identify high-risk existing customers (**collection models**)
- Utilized **Python** on daily basis for data cleaning, feature engineering, and model creation, and monitored model metrics as requested by users (editor: Jupyterhub, Jupyter Notebook, VS Code)
- Developed local dashboards using **dashboarding python packages** such as **Streamlit** and Voila to present results and insights to the internal (Digital Transformation and Business Analyst) team
- Utilized **DataRobot** as an automated machine learning tool when required, **collaborating with a third-party** vendor for the implementation & integration of this software
- Developed & maintained dashboard projects in **Metabase**, which provides critical business insights (Target and Actual, Sales Efficiency) to the BODs & Marketing GMs, built with advanced queries
- Developed & maintained credit scoring project's **Docker & FastAPI** frameworks
- Demonstrated proficiency in **Microsoft SQL** & familiarity with **Postgres** for daily database operations

SMART CITY & COMMUNITY INNOVATION CENTER ITB (SCCIC ITB)

An academic–industry research center at ITB focused on smart city solutions, circular green economy, and innovation ecosystem development through workshops, startup competitions, and sustainable urban research.

Data Scientist

08/21 – 05/22

- Conducted gap analysis & developed a **churn prediction model** for Jasa Raharja that classified insurance arrears and predicted whether someone would be late to pay SWDKLLJ. This project involved working with approximately 6.5 million data points & utilizing the **CRISP-DM** methodology
- Provided assistance to a doctoral student on a paper related to anomaly detection and forecasting in IoT smart-farming sensors using the GRU auto-encoder algorithm in **TensorFlow (neural networks)**

Side Projects

AI Engineer, Chatbot & Agent Development, EventCHI (2024)

A Dutch-based global event-tech startup offering AI-and blockchain-powered event platform services—ranging from cashless NFC payments and NFT ticketing to real-time analytics via its CHI app suite.

- Develop, propose and discuss **Chatbot Architecture** with Team
- Developing chatbot using **Agentic Frameworks** such as **langgraph**, **autogen**, **crewAI**
- Handle all deployment of AI side on **Google Cloud Platform (Cloud Run, Kubernetes Engine, CDN on Google Cloud Storage) & Microsoft Azure (OpenAI models)**
- Develop API using HTTP and **WebSocket Architecture** & Handle the **CI/CD** for AI Chatbot Product
- Develop simple **front end** (HTML-CSS-JS) script for **chatbot demonstration**
- **Flux image generation** prompting and integration
- Deploy **image background removal** technology using open source packages like **rembg** and **PIL**
- Managed **Redis** database operations for seamless & high-performance chat data storage
- Develop **asynchronous multi-image load-save in python code** that reducing the uploading time 90%

Geo-Data Scientist, Route Optimization & Territory Clustering — FMCG & Banking (2023)

- Formulated & solved **Travelling Salesman Problem** for sales-visit sequencing & applied unsupervised clustering to regroup territories across 1M+ rows; achieved 15s per-salesman routing, ~**3hrs nationwide**
- **Owned end-to-end delivery**: algorithm, Streamlit, MongoDB, CI/CD (GitHub, Docker), Google Maps API & OSRM open-source routing - MVP in 2 weeks

Data Related Activities

AI Planet

RAG and Agents Bootcamp

2024

- Vector databases operation with **chromadb**
- Build RAG framework with **llamaIndex**, **langchain**, **beyondllm**
- Build agentic chatbot with **autogen**, **crewAI**, **openAGI**

aiplanet.com/bootcamp/certificate/download/102ac4f6-9516-420a-972b-698616b71bdc/

Data Science (**Big Three** Accuracy in the final assignment as a **datathon**) (leaderboard id: alvinrach)

2021

aiplanet.com/challenges/74/getting-started-with-data-science-bootcamp-final-assignment74/leaderboard/practice

Deep Learning (**First Place** in the final assignment as a **datathon**)

2021

aiplanet.com/challenges/144/gender-determination-by-morphometry-of-eyes-144/leaderboard/practice

Indosat Ooredoo Camp (Idcamp) & FGA Digital Talent Scholarship

2021

Machine Learning (Expert Stage, Decoding ML 5) by Indosat Ooredoo and Ministry of Communication & IT

Self Projects

Walmart Stock Prediction with ~0.45% MAE error (Forecasting Problem using Neural Network)

github.com/alvinrach/25-Time-Series-Stock-Prediction

Car Insurance Sales Prediction (3rd accuracy on Aiplanet bootcamp leaderboard's Notebook)

github.com/alvinrach/21-Car-Insurance-Sales-Prediction

Buy Now Pay Later (Fintech) Prediction

github.com/alvinrach/financial-tech-prediction

Education

BANDUNG INSTITUTE OF TECHNOLOGY

Bandung, Indonesia

Bachelor of Engineering — Chemical Engineering

08/15 – 10/19

- Relevant Courseworks : Engineering Statistics (also as Tutor), Chemical Process Computation, Introduction to Information Technology